



RDFS Semantics and OWL 2 Profiles

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Before we start...

Contact

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GitHub Repository

`https://github.com/city-knowledge-graphs/kg-course-valgrai`

Course Organization

1. Introduction to Knowledge Graphs.
2. Modelling Ontologies with OWL.
3. **RDFS Semantics, Reasoning and OWL 2 Profiles.**
4. From Tabular Data to Knowledge Graphs.
5. Querying Knowledge Graphs and Graph Database Solutions.
6. Knowledge Graphs and Language Models.

Learning outcomes for today

- Familiarization with how reasoning works in OWL and its implications.
- Review of RDFS inference rules.
- Prove entailments via inference rules.
- Critical assessment of the roles of the OWL 2 Profiles
 - Specially focus on OWL 2 RL.
- Application of reasoning in Protégé and programmatically.

Recap

Recap: RDF Example

London is a city in England called Londres in Spanish

```
dbr:london a dbo:City .
```

```
dbr:london dbo:locationCountry dbr:england .
```

```
dbr:london rdfs:label "Londres"@es .
```

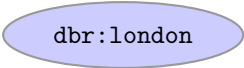

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dbr:london

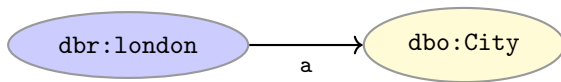
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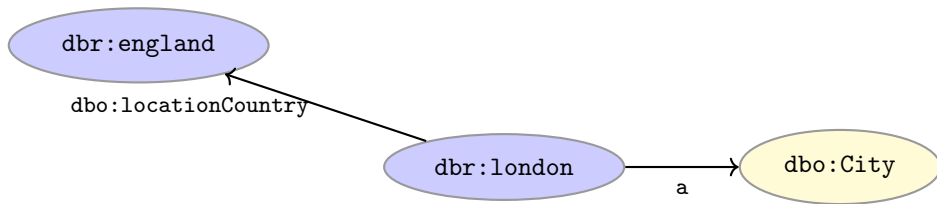
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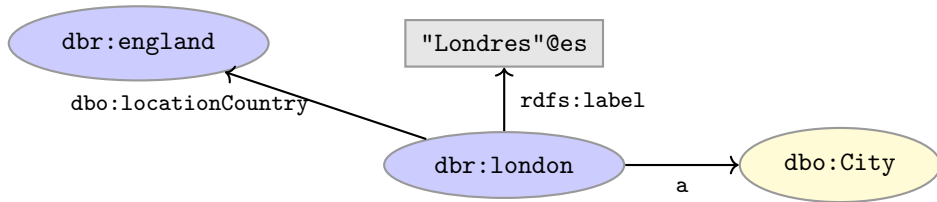
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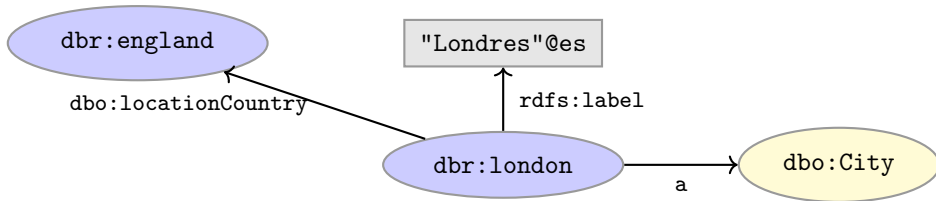
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Sneak peek to the future: SPARQL Example (i)

Return all Cities:

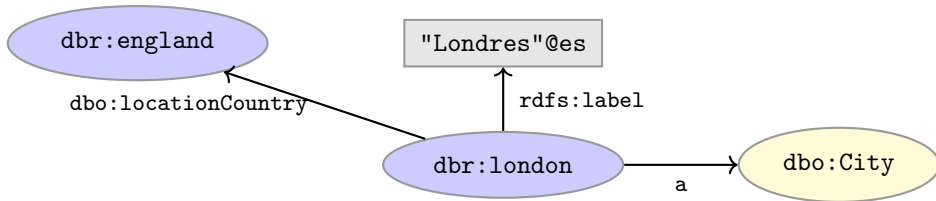
```
PREFIX dbo: <http://dbpedia.org/ontology/>  
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>  
SELECT DISTINCT ?city WHERE {  
    ?city rdf:type dbo:City .  
}
```



Sneak peek to the future: SPARQL Example (i)

Return all Cities: **Query Result= {dbr:london}**

```
PREFIX dbo: <http://dbpedia.org/ontology/>
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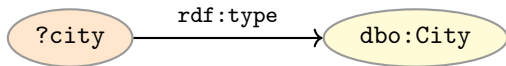


Sneak peek to the future: SPARQL Example (ii)

Return all Cities: **Query Result= {dbr:london}**

```
PREFIX dbo: <http://dbpedia.org/ontology/>
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SELECT DISTINCT ?city WHERE {
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}
```

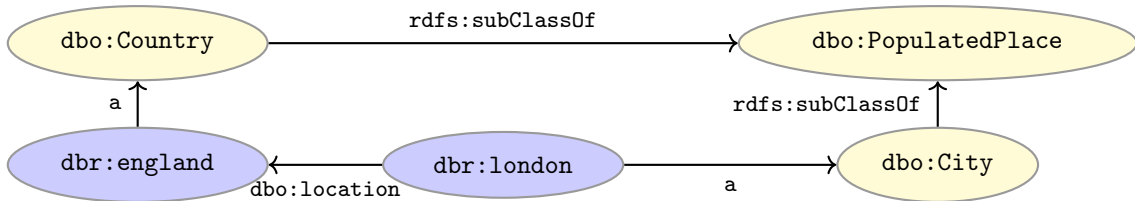
The previous SPARQL query as a graph pattern or set of triple patterns (templates of incomplete triples):



Sneak peek to the future: SPARQL Example (iii)

Return all Populated Places:

```
PREFIX dbo: <http://dbpedia.org/ontology/>  
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>  
SELECT DISTINCT ?place WHERE {  
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}
```



Sneak peek to the future: SPARQL Example (iii)

Return all Populated Places: Query Result= {}

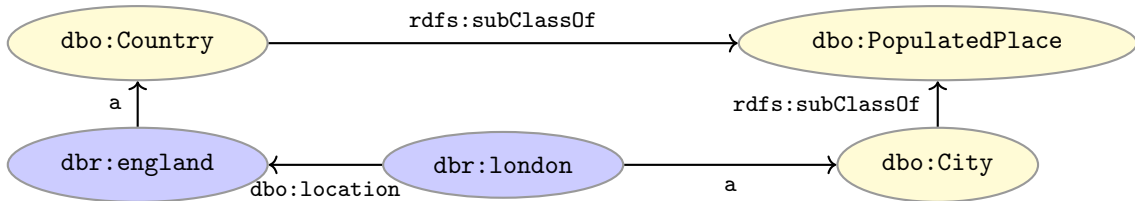
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PREFIX dbo: <http://dbpedia.org/ontology/>
```

```
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```

```
SELECT DISTINCT ?place WHERE {
```

```
    ?place rdf:type dbo:PopulatedPlace .
```

```
}
```



Recap: RDF Grammar for triples

- RDF imposes a basic grammar. A triple consists of *subject*, *predicate*, and *object*

- URI references may occur in all positions
- Literals may only occur in object position
- Blank nodes can not occur in predicate position

s	p	o
✓	✓	✓
✗	✗	✓
✓	✗	✓

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- But one could still define:

```
dbr:london rdf:type "some string"^^xsd:string .
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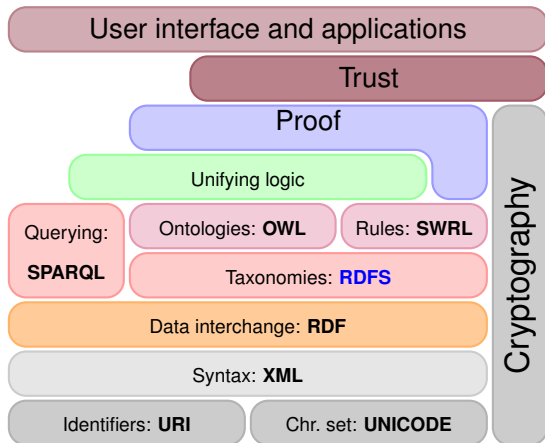
- But one could still define:

`dbr:london rdf:type "some string"^^xsd:string .`

- RDF Schema (RDFS) extends the **grammar** for the “**expected**” triples (*e.g.*, `rdf:type rdf:range rdfs:Resource .`), extends the vocabulary, and include a set of **inference rules**.

RDF Schema (RDFS)

Semantic Web Technology Stack



RDF Schema

- RDF Schema (RDFS) is a vocabulary defined by W3C.
 - <https://www.w3.org/TR/rdf-schema/>
 - <https://www.w3.org/TR/rdf11-nt/>
- Namespace:
`rdfs: http://www.w3.org/2000/01/rdf-schema#`
- Originally designed as a “schema language” like XML Schema.
 - Not strictly – doesn’t describe “valid” RDF graphs.
- A very simple **modeling language** for RDF data → Taxonomies
- **(For our purposes)** it can be seen as a language less expressive than OWL, useful to understand how reasoning works.

RDFS Semantics

- RDFS is a **semantic extension** (adds semantics/meaning) that
 - proposes some **syntactic conditions** on RDF graphs,
 - comes with some (non-ambiguous) **inference rules**, and
 - includes some **(default axiomatic) triples** as part of the specification.

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 - proposes some **syntactic conditions** on RDF graphs,
 - comes with some (non-ambiguous) **inference rules**, and
 - includes some (**default axiomatic**) **triples** as part of the specification.
- For example, RDFS expects as range of `rdf:type` a resource/IRI (e.g., axiomatic triple: `rdf:type rdf:range rdfs:Resource .`)
 - `dbr:london rdf:type "some string"^^xsd:string .`
 - *RDFS*: **Not expected triple**, and will lead to undesired consequences.
 - *OWL*: a **prohibited triple** will lead to an error (validation).

RDFS Vocabulary

- RDFS adds the concept of “classes” which are *sets* of resources.

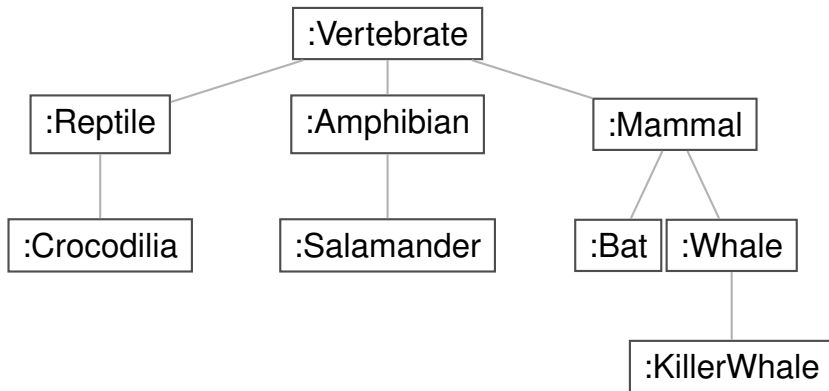
RDFS Vocabulary

- RDFS adds the concept of “classes” which are *sets* of resources.
- Defined resources:
 - `rdfs:Resource`: The class of resources, everything.
 - `rdfs:Class`: The class of classes. (similar to `owl:Class`)
 - `rdfs:Literal`: The class of all literal values.
 - `rdfs:Datatype`: The class of all datatypes.

RDFS Vocabulary

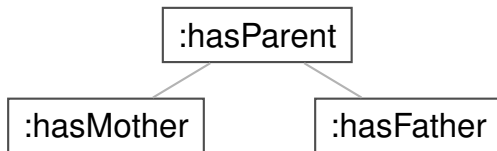
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 - `rdfs:Class`: The class of classes. (similar to `owl:Class`)
 - `rdfs:Literal`: The class of all literal values.
 - `rdfs:Datatype`: The class of all datatypes.
- Defined properties:
 - `rdfs:domain`: The domain (sources) of a relation.
 - `rdfs:range`: The range (targets) of a relation.
 - `rdfs:subClassOf`: Class inclusion.
 - `rdfs:subPropertyOf`: Property inclusion.

Class Taxonomy (via `rdfs:subClassOf`)



(*) As in OWL.

Property Taxonomy (via `rdfs:subPropertyOf`)



(*) As in OWL.

Expected RDF/RDFS resources

Types of resources or elements:

- *Object Properties* like `foaf:knows`
- *Datatype Properties* like `dc:title`, `foaf:name`
- *Classes* like `foaf:Person`
- *Built-ins*, a fixed set including `rdf:type`, `rdfs:domain`, etc.
- *Individuals* (all the rest, “usual” resources) like `city:ernesto`
- *Datatypes* like `xsd:integer`
- *Literals* like `"ernesto"`, `"39"`

(*) Not real split of properties into object and data properties in RDFS. This comes in OWL.

Expected RDF/RDFS triple grammar

Triples

indi o-prop indi .

indi d-prop "lit" .

indi rdf:type class .

class rdfs:subClassOf class .

o-prop rdfs:subPropertyOf o-prop .

d-prop rdfs:subPropertyOf d-prop .

o-prop rdfs:domain class .

o-prop rdfs:range class .

d-prop rdfs:domain class .

d-prop rdfs:range datatype .

(Default) RDFS axiomatic triples (excerpt)

- Indeed RDF and RDFS include a set of default triples to guide the above grammar of expected triples.
- Only resources have types:
`rdf:type rdfs:domain rdfs:Resource .`
- Types are classes:
`rdf:type rdfs:range rdfs:Class .`
- Ranges apply only to properties:
`rdfs:range rdfs:domain rdf:Property .`

RDFS Entailment Rules: https://www.w3.org/TR/rdf-mt/#rdfs_interp

(Default) RDFS axiomatic triples (excerpt)

- Ranges are classes:

```
rdfs:range rdfs:range rdfs:Class .
```

- Only properties have subproperties:

```
rdfs:subPropertyOf rdfs:domain rdf:Property .
```

- Only classes have subclasses:

```
rdfs:subClassOf rdfs:domain rdfs:Class .
```

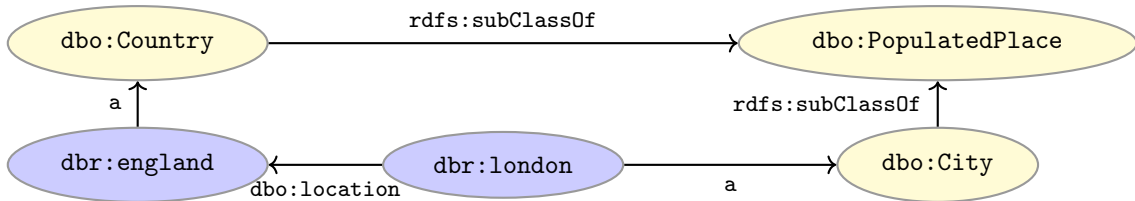
- ... (another 30 or so)

Entailment via Model-Theoretic Semantics

SPARQL Example

Return all Populated Places:

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PREFIX dbo: <http://dbpedia.org/ontology/>
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SELECT DISTINCT ?place WHERE {
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SPARQL Example

Return all Populated Places: Query Result= {}

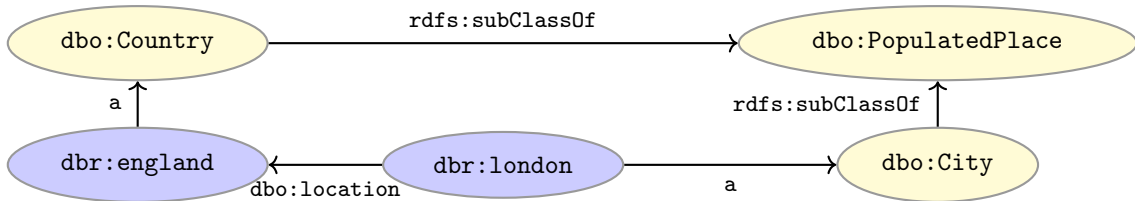
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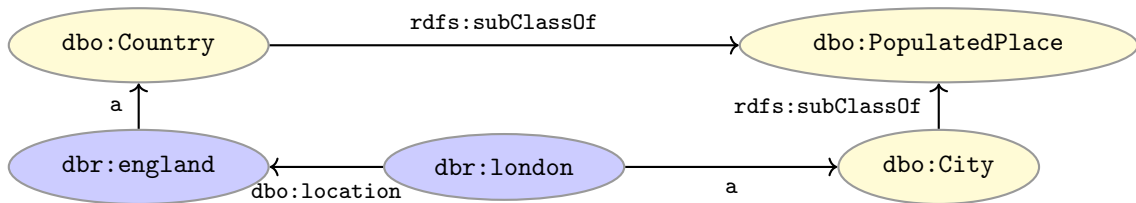
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```

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}
```



Entailment in RDFS

- Given a set of triples $\mathcal{G}$ (i.e., a Graph) can we entail a triple t ($\mathcal{G} \models t$)?
- Can we entail the triple: `dbr:london rdfs:type dbo:PopulatedPlace` and add it to the graph below? (*Graph expansion via reasoning*).
- Similarly for `dbr:england`



Model-Theoretic Semantics (i)

- **Interpretations** might be conceived as potential "realities" or "worlds".
- Interpretations assign values to elements.
 - (*The **intuitions** behind set-theory are **formally represented**.*)

Model-Theoretic Semantics (i)

- **Interpretations** might be conceived as potential "realities" or "worlds".
- Interpretations assign values to elements.
 - (The ***intuitions*** behind set-theory are ***formally represented.***)
- Given an interpretation $\mathcal{I}$ and a set of triples $\mathcal{G}$
- $\mathcal{G}$ is valid in $\mathcal{I}$ (written $\mathcal{I} \models \mathcal{G}$), iff $\mathcal{I} \models t$ for all $t \in \mathcal{G}$.
- Then $\mathcal{I}$ is also called a **model** of $\mathcal{G}$.

Model-Theoretic Semantics (ii)

- The following interpretation $\mathcal{I}_1$ is a model of our example $\mathcal{G}$:
 - $\text{dbo:City}^{\mathcal{I}_1} = \{\text{dbr:london}^{\mathcal{I}_1}\}$
 - $\text{dbo:Country}^{\mathcal{I}_1} = \{\text{dbr:england}^{\mathcal{I}_1}\}$
 - $\text{dbo:PopulatedPlace}^{\mathcal{I}_1} = \{\text{dbr:london}^{\mathcal{I}_1}, \text{dbr:england}^{\mathcal{I}_1}\}$
 - $\text{dbo:location}^{\mathcal{I}_1} = \{\langle \text{dbr:london}^{\mathcal{I}_1}, \text{dbr:england}^{\mathcal{I}_1} \rangle\}$

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 - $\text{dbo:location}^{\mathcal{I}_1} = \{\langle \text{dbr:london}^{\mathcal{I}_1}, \text{dbr:england}^{\mathcal{I}_1} \rangle\}$
- $\mathcal{I}_1 \models \mathcal{G}$ (is a model of $\mathcal{G}$) as the following holds:
 - $\text{dbo:City}^{\mathcal{I}_1} \subseteq \text{dbo:PopulatedPlace}^{\mathcal{I}_1}$
 - $\text{dbo:Country}^{\mathcal{I}_1} \subseteq \text{dbo:PopulatedPlace}^{\mathcal{I}_1}$
 - $\text{dbr:london}^{\mathcal{I}_1} \in \text{dbo:City}^{\mathcal{I}_1}$ and $\text{dbr:england}^{\mathcal{I}_1} \in \text{dbo:Country}^{\mathcal{I}_1}$
 - $\langle \text{dbr:london}^{\mathcal{I}_1}, \text{dbr:england}^{\mathcal{I}_1} \rangle \in \text{dbo:location}^{\mathcal{I}_1}$

Model-Theoretic Semantics: Exercise

Think about another interpretation $\mathcal{I}_2$ such that $\mathcal{I}_2 \models \mathcal{G}$ (i.e., is a model of $\mathcal{G}$). $\mathcal{G}$ being our example as below:

- $\text{dbo:City}^{\mathcal{I}_2} \subseteq \text{dbo:PopulatedPlace}^{\mathcal{I}_2}$
- $\text{dbo:Country}^{\mathcal{I}_2} \subseteq \text{dbo:PopulatedPlace}^{\mathcal{I}_2}$
- $\text{dbr:london}^{\mathcal{I}_2} \in \text{dbo:City}^{\mathcal{I}_2}$ and $\text{dbr:england}^{\mathcal{I}_2} \in \text{dbo:Country}^{\mathcal{I}_2}$
- $\langle \text{dbr:london}^{\mathcal{I}_2}, \text{dbr:england}^{\mathcal{I}_2} \rangle \in \text{dbo:location}^{\mathcal{I}_2}$

Model-Theoretic Semantics: Exercise Solution

- The following interpretation $\mathcal{I}_2$ is a model of our example $\mathcal{G}$:
 - $\text{dbo:City}^{\mathcal{I}_2} = \{\text{dbr:london}^{\mathcal{I}_2}, \text{dbr:berlin}^{\mathcal{I}_2}\}$
 - $\text{dbo:Country}^{\mathcal{I}_2} = \{\text{dbr:england}^{\mathcal{I}_2}\}$
 - $\text{dbo:PopulatedPlace}^{\mathcal{I}_2} = \{\text{dbr:london}^{\mathcal{I}_2}, \text{dbr:england}^{\mathcal{I}_2}, \text{dbr:berlin}^{\mathcal{I}_2}\}$
 - $\text{dbo:location}^{\mathcal{I}_2} = \{\langle \text{dbr:london}^{\mathcal{I}_2}, \text{dbr:england}^{\mathcal{I}_2} \rangle\}$

Model-Theoretic Semantics: Exercise Solution

- The following interpretation $\mathcal{I}_2$ is a model of our example $\mathcal{G}$:
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 - $\text{dbr:london}^{\mathcal{I}_2} \in \text{dbo:City}^{\mathcal{I}_2}$ and $\text{dbr:england}^{\mathcal{I}_2} \in \text{dbo:Country}^{\mathcal{I}_2}$
 - $\langle \text{dbr:london}^{\mathcal{I}_2}, \text{dbr:england}^{\mathcal{I}_2} \rangle \in \text{dbo:location}^{\mathcal{I}_2}$

Model-Theoretic Semantics: Exercise (ii)

- $t = \text{dbr:london rdf:type dbo:PopulatedPlace}$
- Does $\mathcal{I}_1 \models t$? and $\mathcal{I}_2 \models t$?

Model-Theoretic Semantics: Exercise (ii)

- $t = \text{dbr:london rdf:type dbo:PopulatedPlace}$
- Does $\mathcal{I}_1 \models t$? and $\mathcal{I}_2 \models t$?
 - **Yes:** $\text{dbo:PopulatedPlace}^{\mathcal{I}_1} = \{\text{dbr:london}, \text{dbr:england}\}$, that is,
 $\text{dbr:london} \in \text{dbo:PopulatedPlace}^{\mathcal{I}_1}$

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- Does $\mathcal{G} \models t$?

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- Does $\mathcal{G} \models t$?
 - **if and only if**
 - For **any interpretation** $\mathcal{I}$ with $\mathcal{I} \models \mathcal{G}$
 - $\mathcal{I} \models t$.
 - (Yes, in this case too.)

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 - (Yes, in this case too.)
- Does $\mathcal{G} \models t_2$ ($t_2 = \text{dbr:london rdf:type dbo:Country}$)?

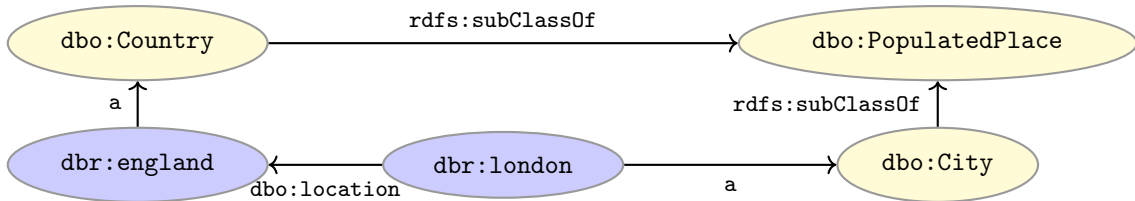
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 - **if and only if**
 - For **any interpretation** $\mathcal{I}$ with $\mathcal{I} \models \mathcal{G}$
 - $\mathcal{I} \models t$.
 - (Yes, in this case too.)
- Does $\mathcal{G} \models t_2$ ($t_2 = \text{dbr:london rdf:type dbo:Country}$)?
 - **No:** Our $\mathcal{I}_1$ and $\mathcal{I}_2$ are counter examples. *e.g.*, $\mathcal{I}_1 \models \mathcal{G}$ but $\mathcal{I}_1 \not\models t_2$

SPARQL Example: with entailment

Return all Populated places:

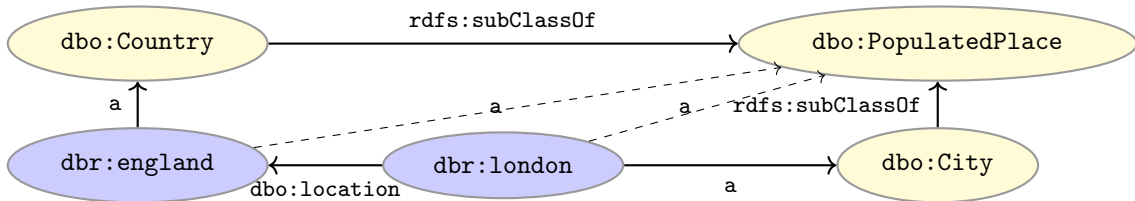
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SPARQL Example: with entailment

Return all Populated places: **Query Result= {dbr:england, dbr:london}**

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SELECT DISTINCT ?place WHERE {
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Model-Theoretic Semantics in practice

- Model-theoretic semantics yields an unambiguous notion of entailment.
- In principle, **all interpretations** need to be considered.
- However there are **infinitely many** such interpretations,
- An **algorithm should terminate** in finite time.

Foundations of Semantic Web Technologies. Chapter 3.

Entailment via Inference Rules

Syntactic Reasoning

- From the computation point of view, we need means to decide **entailment syntactically**.
- Syntactic methods operate
 - only on the form of a statement, that is on its **concrete grammatical structure** (*i.e.*, triples),
 - without recurring to interpretations.
- Syntactic methods should justify that their so-called **operational semantics** are expected with respect to model-theoretic semantics.

Inference rules (i)

- Inference rules (also known as deduction rules or derivation rules) is an option to **describe syntactic solutions**.
- The general form of an inference rule is:

$$\frac{P_1, \dots, P_n}{P}$$

- the P_i are **premises** (body)
- and P is the **conclusion** (head).
- An inference rule may have,
 - any number of premises (typically one or two),
 - but only one conclusion.

Inference rules (ii)

- Recall that syllogisms (*i.e.*, inference) can be traced back to Aristotle
- Example:

All human are mortal
Socrates is a human

Therefore, Socrates is mortal

Inference rules (iii)

- The whole set of inference rules given for a logic is called **deduction calculus**.
- $\vdash$ is the **inference relation**, while $\models$ was the entailment relation using model theoretic semantics.
 - We write $\Gamma \vdash P$ if we can deduce P from the premises Γ .

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- $\vdash$ is the **inference relation**, while $\models$ was the entailment relation using model theoretic semantics.
 - We write $\Gamma \vdash P$ if we can deduce P from the premises Γ .
- **In our setting:**
 - the **premises** Γ are a **set of triples** (*i.e.*, a (sub)graph $\mathcal{G}$),
 - the **conclusion** P is a **new triple** t
 - After applying the rules to $\mathcal{G}$ we will get an **expanded graph** $\mathcal{G}'$

RDFS Inference Rules

RDFS supports several rules. Organized into three groups:

1. **Type propagation:**

- “London is a City, all Cities are populated places, so...”

2. **Property propagation:**

- “London is the capital of England, anything that is capital of a country is also located in that country, so...”

3. **Domain and range propagation:**

- “Everything that has a capital is a country, so England is a...”
- “Everything that is a capital is a city, so London is a...”

RDFS Entailment Rules: <https://www.w3.org/TR/rdf-mt/#RDFSRules>

Type propagation

- **Members of superclasses:**

$$\frac{A \text{ rdfs:subClassOf } B . \quad x \text{ rdf:type } A .}{x \text{ rdf:type } B .} \text{ rdfs9}$$

(*) rdfs9, rdfs10, rdfs11 are the names of the inference rules in the W3C standard.
A, B are classes; x is an instance.

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$$\frac{A \text{ rdfs:subClassOf } B . \quad B \text{ rdfs:subClassOf } C .}{A \text{ rdfs:subClassOf } C .} \text{ rdfs11}$$

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Type propagation: Examples

- **Members of superclasses:**

$$\frac{:City \text{ rdfs:subClassOf } :PopulatedPlace . \quad :london \text{ rdf:type } :City .}{:london \text{ rdf:type } :PopulatedPlace .} \text{ rdfs9}$$

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$$\frac{:City \text{ rdf:type } \text{ rdfs:Class } .}{:City \text{ rdfs:subClassOf } :City .} \text{ rdfs10}$$

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$$\frac{:City \text{ rdfs:subClassOf } :PopulatedPlace . \quad :PopulatedPlace \text{ rdfs:subClassOf } :Place .}{:City \text{ rdfs:subClassOf } :Place .} \text{ rdfs11}$$

Property Propagation

– Transitivity:

$$\frac{P \text{ rdfs:subPropertyOf } Q . \quad Q \text{ rdfs:subPropertyOf } R .}{P \text{ rdfs:subPropertyOf } R .} \text{ rdfs5}$$

(*) P, Q are properties; u, v are instances.

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– Reflexivity:

$$\frac{P \text{ rdf:type } \text{rdf:Property} .}{P \text{ rdfs:subPropertyOf } P .} \text{ rdfs6}$$

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- **Property transfer:**

$$\frac{P \text{ rdfs:subPropertyOf } Q . \quad u P v .}{u Q v .} \text{ rdfs7}$$

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Property Propagation: Examples

– Transitivity:

$$\frac{\text{ :has_writer rdfs:subPropertyOf :has_author . } \quad \text{ :has_author rdfs:subPropertyOf :has_creator . }}{\text{ :has_writer rdfs:subPropertyOf :has_creator . }} \text{ rdfs5}$$

– Reflexivity:

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– Property transfer:

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Domain and range propagation

Typing triggered by the use of properties.

– **Domain propagation:**

$$\frac{P \text{ rdfs:domain } A . \quad x P y .}{x \text{ rdf:type } A .} \text{ rdfs2}$$

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Domain and range propagation

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- **Domain propagation:**

$$\frac{P \text{ rdfs:domain } A . \quad x P y .}{x \text{ rdf:type } A .} \text{ rdfs2}$$

- **Range propagation:**

$$\frac{P \text{ rdfs:range } B . \quad x P y .}{y \text{ rdf:type } B .} \text{ rdfs3}$$

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Domain and Range Propagation: Examples

– Domain propagation:

:capitalOf rdfs:domain :City . :london :capitalOf :england .
:london rdf:type :City . rdfs2

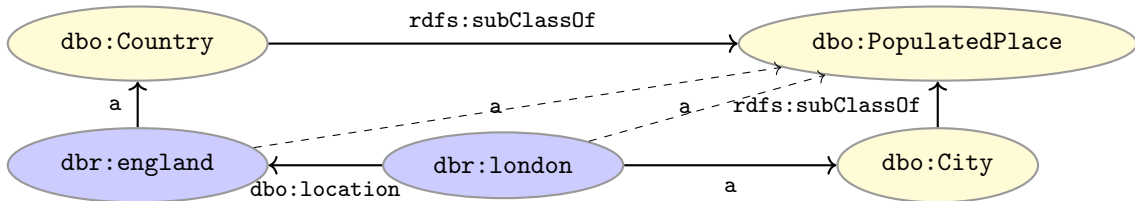
– Range propagation:

:capitalOf rdfs:range :Country . :london :capitalOf :england .
:england rdf:type :Country . rdfs3

SPARQL Example: with entailment

Return all Populated places: **Query Result= {dbr:england, dbr:london}**

```
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?place WHERE {
    ?place rdf:type dbo:PopulatedPlace .
}
```



Characteristics of RDFS Semantics

Entailment and Inference in RDFS

- Both have the **monotonic** property.
 - If a graph $\mathcal{G} \models t$ (or $\mathcal{G} \vdash t$),
 - then adding more triples (*e.g.*, t_1) does not alter the entailment $\mathcal{G} \cup \{t_1\} \models t$ (or derivation $\mathcal{G} \cup \{t_1\} \vdash t$)

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- The set of RDFS rules we have seen are **sound**.
 - If $\mathcal{G} \vdash t$ then $\mathcal{G} \models t$
- But **not complete**.
 - Not always applies that If $\mathcal{G} \models t$ then $\mathcal{G} \vdash t$
 - Due to how RDFS deals with datatypes.

(Non) Validation in RDFS (i)

- RDFS was conceived of as a schema language for RDF
- However, the statements in an RDFS graph **never trigger inconsistencies**.
- Reasoning will not lead to a “contradiction”, “error”, “non-valid document”
- Inference rules in RDFS add more triples, but **do not detect errors**.

(Non) Validation in RDFS (ii)

- RDFS has **no notion of negation**

- For instance, the two triples

`dbr:london rdf:type dbo:Country .`

`dbr:london rdf:type dbo:City .`

are not inconsistent.

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- There is also **not clear notion of disjointness** among RDF resources:
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are not inconsistent.
- There is also **not clear notion of disjointness** among RDF resources: Object Properties, Datatype Properties, Classes, Built-in properties, Individuals, Datatypes and Literals.
- **OWL** (and OWL 2) includes additional vocabulary and also **consistency-checks**.

OWL 2 Reasoning and Profiles

Recap: OWL (The Web Ontology Language)

- A **W3C** recommendation:

- OWL 1 (2004): <http://www.w3.org/TR/owl-ref/>

- OWL 2 (2009): <https://www.w3.org/TR/owl2-overview/>

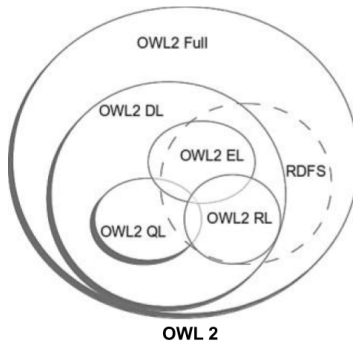
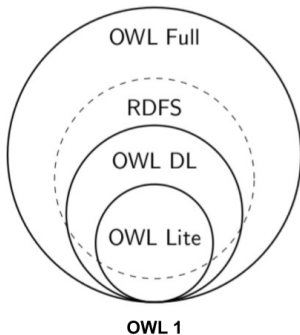


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- A **W3C recommendation**:
 - OWL 1 (2004): <http://www.w3.org/TR/owl-ref/>
 - OWL 2 (2009): <https://www.w3.org/TR/owl2-overview/>
- OWL semantics based on **Description Logics (DL)**.
 - Family of knowledge representation languages
 - Decidable subset of First Order logic (FOL)
 - Original called: **Terminological language** or **concept language**

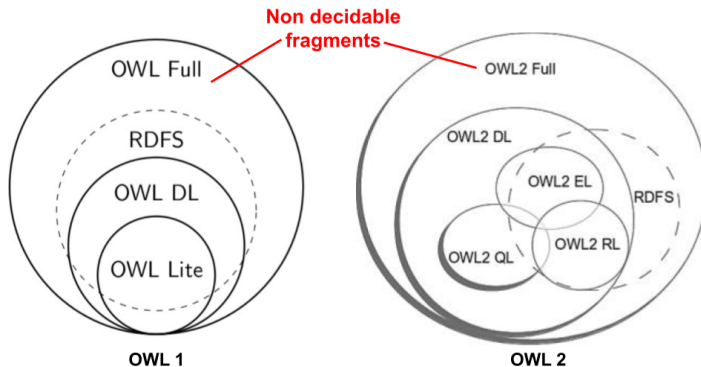


Recap: OWL 1, OWL 2 (profiles) and RDFS



Olivier Cure and Guillaume Blin. RDF Database Systems (Chapter 3). 2015. Elsevier.

Recap: OWL 1, OWL 2 (profiles) and RDFS



† **Reasoning in OWL 2** will partially get the same consequences as in the RDFS inference rules and many more.

Automated Reasoning in OWL

- Formal semantics allows the automatic deduction of **new facts** (*i.e.*, *not explicit but derived from others*).
- Also allows us to perform checks that aim to detect the **correctness** of the designed model (*e.g.*, is `dbr:london` a `:dbo:Country`?). **New respect to RDFS.**

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- In the form of **logical errors**:
 - `dbo:City` and `dbo:Country` are disjoint classes.
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- Extremely valuable for designing **correct** ontologies/KGs.

OWL 2 Reasoning

Recap: OWL 2 Axioms into Boxes

- Traditionally OWL 2 axioms are put in boxes.
- The **TBox** (terminological knowledge)
 - Typically independent of any actual instance data.
 - Property axioms are also referred to as **RBox**
- The **ABox** (assertional knowledge)
 - Contains facts about concrete instance.

OWL 2 TBox Reasoning

(Standard) Reasoning tasks that use only the TBox $\mathcal{T}^\dagger$

- Concept **unsatisfiability**: Given C , does $\mathcal{T} \models C \sqsubseteq \perp$? (i.e., $C^{\mathcal{I}} = \emptyset$)

† (**Model-Theoretic Semantics**) The answer to ‘does $\mathcal{T} \models \alpha$?’ will be positive if for each interpretation $\mathcal{I}$ such that $\mathcal{I} \models \mathcal{T}$, $\mathcal{I} \models \alpha$ too.

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- Conjunctive Query Answering (**SPARQL**).

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OWL 2 Reasoning Algorithms

- Reasoning in OWL 2 is typically based on **(Hyper)Tableau Reasoning Algorithms** (tableau = truth tree)
- Reasoning tasks **reduced to (un)satisfiability**.
- Algorithm tries to construct an **abstraction** of a model.

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Seminars by Prof. Ian Horrocks: <http://www.cs.ox.ac.uk/people/ian.horrocks/Seminars/seminars.html>

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- Reasoning tasks **reduced to (un)satisfiability**.
- Algorithm tries to construct an **abstraction** of a model.
- State-of-the-art algorithms:
 - *e.g.*, **HermiT** (default option in Protégé).
 - Implement a number of (search) **optimisations**.
 - **Effective** with many realistic ontologies

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Tractability Problems with OWL 2 Reasoning

- Problems with **very large** and/or **cyclical ontologies**.
 - Ontologies may define hundred of thousands of terms (*e.g.*, SNOMED CT)
 - Large number of tests for classification (each test can lead to the construction of very large models).

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- Problems with **medium/large** data sets (ABoxes)
 - OWL 2 Reasoners typically optimized for TBox reasoning tasks.
 - Data also brings additional complexity.

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OWL 2 profiles

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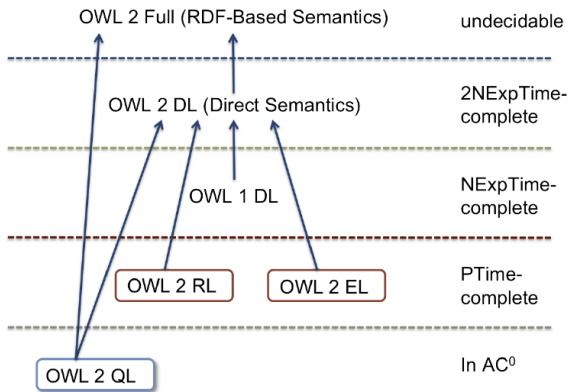
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 - **OWL 2 EL**:
 - A lightweight language with polynomial time reasoning.
 - **OWL 2 RL**:
 - Designed for compatibility with rule-based inference tools.
 - Efficient reasoning with large datasets.

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Data Complexity OWL 2 Profiles



<https://www.w3.org/TR/owl2-profiles/>

OWL EL profile (i)

Based on the DL $\mathcal{EL}^{++}$. Concept descriptions (simplified)

$C, D \rightarrow$	A		(atomic concept)
	$\top$		(universal concept)
	$\perp$		(bottom concept)
	$\{a\}$		(<i>singular</i> enumeration)
	$C \sqcap D$		(intersection)
	$\exists R.C$		(existential restriction)

Axioms

- $C \sqsubseteq D$ and $C \equiv D$ for concept descriptions D and C .
- $P \sqsubseteq Q$ and $P \equiv Q$ for roles P, Q . Also Domain and Range.
- $C(a)$ and $R(a, b)$ for concept C , role R and individuals a, b .

OWL EL Profile (ii)

- ✓ Standard reasoning tasks in P time
- ✓ Very good for large ontologies.
- ✓ Used in many biomedical ontologies (*e.g.*, SNOMED CT).

Not supported features, simplified:

- ✗ negation (but $C \sqcap D \sqsubseteq \perp$ possible)
- ✗ disjunction
- ✗ universal quantification and cardinalities
- ✗ inverse roles and some role characteristics
- ✗ reduced list of datatypes

OWL EL Profile (iii)

- Reasoning can be performed via saturation[†] (*i.e.*, inference rules).
- For example:

$$\frac{A \sqsubseteq B \quad B \sqsubseteq C}{A \sqsubseteq C}$$
$$\frac{A \sqsubseteq \exists R.B \quad \exists S.B \sqsubseteq C \quad S \sqsubseteq R}{A \sqsubseteq C}$$

[†] Using a saturation-based approach over an OWL 2 ontology is not possible.

ELK reasoner (also available as Protégé plugin): <https://github.com/liveontologies/elk-reasoner/wiki>

OWL QL Profile (i)

Based on DL-Lite_R. Concept descriptions (simplified)

$C, C' \rightarrow$	A		(atomic concept)
	$\exists R. \top$		(existential restriction with $\top$ only)
$D, D' \rightarrow$	A		(atomic concept)
	$\exists R. D$		(existential restriction)
	$\neg D$		(negation)
	$D \sqcap D'$		(intersection)

Axioms

- $C \sqsubseteq D$ for concept descriptions D and C (and $C \equiv C'$).
- $P \sqsubseteq Q$ and $P \equiv Q$ for roles P, Q . Also Domain and Range.
- $C(a)$ and $R(a, b)$ for concept C , role R and individuals a, b .

OWL QL Profile (ii)

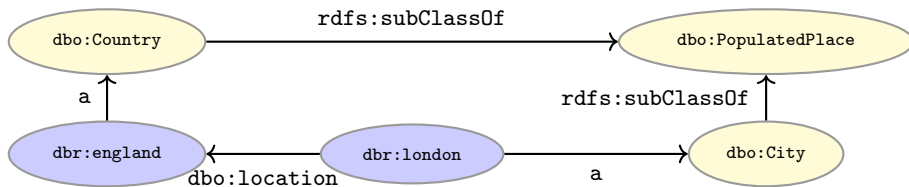
- ✓ Required language so that queries can be rewritten using the TBox.
- ✓ Used in Ontology Based Data Access (OBDA) where SPARQL queries are translated to SQL

Not supported, simplified:

- ✗ disjunction
- ✗ universal quantification, cardinalities, and functional roles
- ✗ `= (SameIndividual)`
- ✗ enumerations (closed classes)
- ✗ subproperties of chains, transitivity
- ✗ reduced list of datatypes

OWL QL Profile (iii)

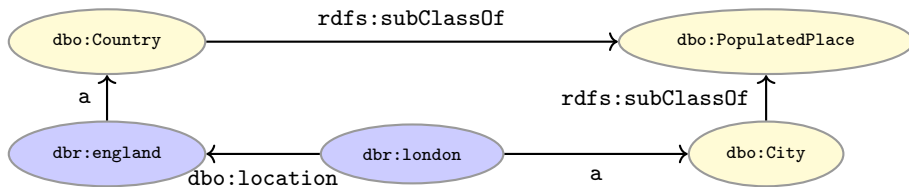
- Reasoning is performed via backward chaining (e.g., rewriting of a given query Q into Q' via the ontology axioms, instead of expanding the graph). For example:



OWL QL Profile (iii)

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Q : `SELECT DISTINCT ?place WHERE {?place rdf:type dbo:PopulatedPlace . }`

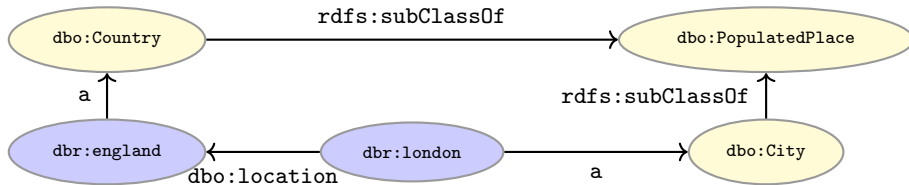


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Q : `SELECT DISTINCT ?place WHERE {?place rdf:type dbo:PopulatedPlace . }`

Q' : `SELECT DISTINCT ?place WHERE {
 {?place rdf:type dbo:PopulatedPlace .}
 UNION {?place rdf:type dbo:Country .}
 UNION {?place rdf:type dbo:City .} }`



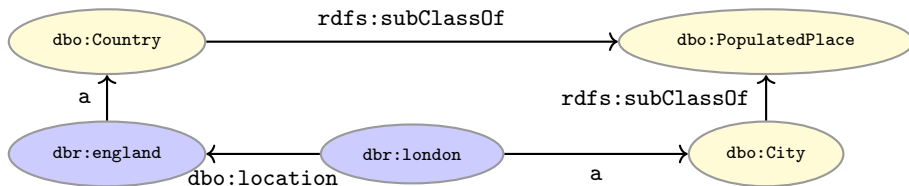
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 UNION {?place rdf:type dbo:Country .}
 UNION {?place rdf:type dbo:City .} }`

Q' Result= {dbr:england, dbr:london}



OWL 2 RL Profile (i)

Based on Description Logic Programs (DLP). Concept descriptions:

$C, C' \rightarrow$	A		(atomic concept)
	$C \sqcap C'$		(intersection)
	$C \sqcup C'$		(union)
	$\exists R.C$		(existential restriction)
$D, D' \rightarrow$	A		(atomic concept)
	$D \sqcap D'$		(intersection)
	$\forall R.D$		(universal restriction)

Axioms

- $C \sqsubseteq D, C \equiv C', \top \sqsubseteq \forall R.D, \top \sqsubseteq \forall R^-.D, R \sqsubseteq P, R \equiv P^-$ and $R \equiv P$ for roles R, P and concept descriptions C, C' and D . Also Domain and Range.
- $C(a)$ and $R(a, b)$ for concept C , role R and individuals a, b .

OWL 2 RL Profile (ii)

- ✗ Puts syntactic constraints in the way in which constructs are used (i.e., syntactic subset of OWL 2).
- ✗ Imposes a reduced list of allowed datatypes
- ✓ OWL 2 RL axioms can be directly translated into datalog rules
- ✓ Enables desirable computational properties using rule-based reasoning engines.

OWL 2 RL Profile (iii)

- Reasoning via full materialisation of the graph, similarly to RDFS inference rules. *e.g.*:

$$\frac{p1 \text{ owl:inverseOf } p2 . \quad ?x ?p1 ?y .}{?y ?p2 ?x .}$$

- See W3C specification for further inference rules in OWL 2 RL.
- After applying the inference rules over $\mathcal{G}$ one gets an expanded graph $\mathcal{G}'$ and queries can be executed over that graph.

W3C: https://www.w3.org/TR/owl2-profiles/#Reasoning_in_OWL_2_RL_and_RDF_Graphs_using_Rules
GraphDB: <https://graphdb.ontotext.com/documentation/standard/reasoning.html>

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OWL 2 RL Profile (iii)

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```
:hasChild owl:inverseOf :hasParent .      :ernesto :hasChild :diego  
:diego :hasParent :ernesto .
```

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Practical Examples: OWL Reasoning

Necessary conditions and primitive classes

Hawaiian pizza **implies** having pineapple as ingredient (among others); but not the other way round. $\text{HawaiianPizza} \sqsubseteq \exists \text{hasIngredient.Pineapple}$

Description: Hawaiian pizza

Equivalent To +

SubClass Of +

'American pizza'	? @ x o
hasIngredient some 'Tomato sauce'	? @ x o
hasIngredient some Cheese	? @ x o
hasIngredient some Ham	? @ x o
hasIngredient some Pineapple	? @ x o
NamedPizza	? @ x o

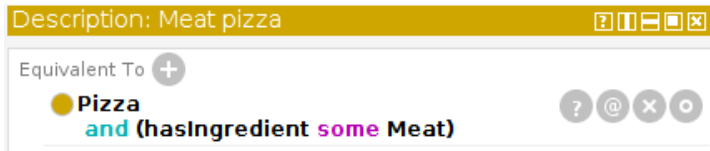
Sufficient conditions and defined classes

$\text{MeatPizza} \equiv \text{Pizza} \sqcap \exists \text{hasIngredient}.\text{Meat}$

Meat pizza **implies** having meat as ingredient (and being pizza).

A pizza with meat as ingredient **implies** being a meat pizza.

Hawaiian pizzas have ham as ingredient and thus they are meat pizzas.



Sufficient conditions: example reasoning in OWL 2 EL

$$\frac{HawaiianPz \sqsubseteq \exists hasIng.Ham \quad \exists hasIng.Meat \sqsubseteq MeatPz \quad Ham \sqsubseteq Meat}{HawaiianPz \sqsubseteq MeatPz}$$

† Using a saturation-based approach over an OWL 2 DL ontology is not possible. But it is a good intuition about how reasoning works when using Protégé with ELK and, to some extent, HermiT.

Detecting modelling errors

- Ice cream **implies** having fruit as topping:
 $\text{IceCream} \sqsubseteq \exists \text{hasTopping.FruitTopping}$
- Ice cream is **disjoint with** Pizza ($\text{IceCream} \sqcap \text{Pizza} \sqsubseteq \perp$)
- The domain of 'has topping' is pizza, that is, having any topping **implies** being a pizza: $\exists \text{hasTopping}.\top \sqsubseteq \text{Pizza}$
- Domain is a type of sufficient condition, global scope for the property.

Description: IceCream

Equivalent To $+$

- owl:Nothing

SubClass Of $+$

- Food
- hasTopping some FruitTopping

Disjoint With $+$

- PizzaTopping, Pizza, PizzaBase

Description: hasTopping

Equivalent To $+$

SubProperty Of $+$

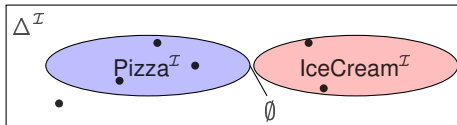
- hasIngredient
- inverse (isIngredientOf)

Inverse Of $+$

- isToppingOf

Domains (intersection) $+$

- Pizza



Detecting modelling errors: reasoning in OWL 2 EL

$$\frac{IceCr \sqcap Pizza \sqsubseteq \perp \quad \exists hasTop.\top \sqsubseteq Pizza \quad IceCr \sqsubseteq \exists hasTop.Fruit}{IceCr \sqsubseteq \perp}$$

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Detecting modelling errors: part-of VS subclass-of (RDFS)

:City rdfs:subClassOf :Country . ?

- **Members of superclasses (rdfs9 rule):**

:City rdfs:subClassOf :Country . :london rdf:type :City . rdfs9

Detecting modelling errors: part-of VS subclass-of (RDFS)

:City rdfs:subClassOf :Country . ? **Incorrect**

- **Members of superclasses (rdfs9 rule):**

:City rdfs:subClassOf :Country . :london rdf:type :City . rdfs9
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Detecting modelling errors: part-of VS subclass-of (RDFS)

:City rdfs:subClassOf :Country . ? **Incorrect**

– Members of superclasses (rdfs9 rule):

:City rdfs:subClassOf :Country . :london rdf:type :City . rdfs9
:london rdf:type :Country .

:City rdfs:subClassOf :locatedIn some :Country

Detecting modelling errors: property hierarchy (RDFS)

`:isLocatedInCity rdfs:subPropertyOf :isLocatedInCountry . ?`

- **Property transfer (rdfs7 rule):**

`:isLocInCity rdfs:subPropertyOf :isLocInCountry .      :big_ben :isLocInCity :london .`

Detecting modelling errors: property hierarchy (RDFS)

`:isLocatedInCity rdfs:subPropertyOf :isLocatedInCountry .` ? **Incorrect**

- Property transfer (rdfs7 rule):

`:isLocInCity rdfs:subPropertyOf :isLocInCountry .      :big_ben :isLocInCity :london .`
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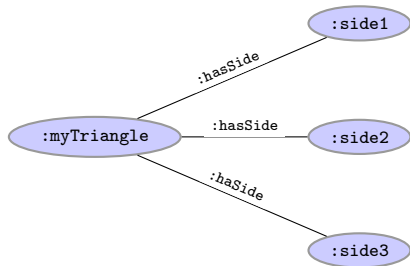
`:isLocInCity rdfs:subPropertyOf :isLocInCountry .      :big_ben :isLocInCity :london .`
`:big_ben :isLocInCountry :london .`

`:isLocatedInCity rdfs:subPropertyOf :isLocatedIn .`

`:isLocatedInCountry rdfs:subPropertyOf :isLocatedIn .`

OWL 2 and Open World Assumption

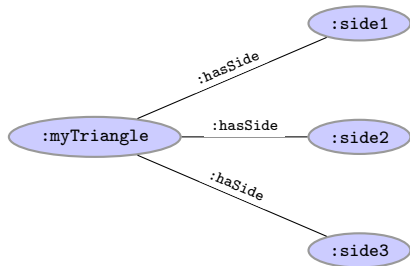
- `:Triangle` EquivalentTo `:hasSide` exactly 3 `:Side`



- is `:myTriangle` a `:Triangle`?

OWL 2 and Open World Assumption

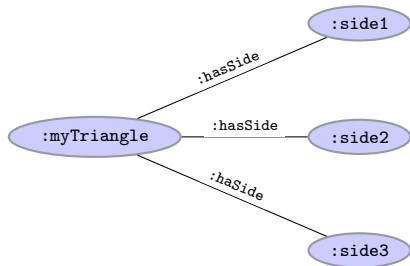
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- is `:myTriangle` a `:Triangle`? **I don't know** because of OWA and NUNA.

OWL 2 and Open World Assumption

- `:Triangle EquivalentTo :hasSide exactly 3 :Side`



- is `:myTriangle` a `:Triangle`? **I don't know** because of OWA and NUNA.
- **Solution:** reasoning in OWL can be complemented with SPARQL queries (in this case with aggregates) → SPARQL 1.1

Laboratory: RDFS Semantics and OWL 2 RL

Tasks

- **Manually checking** inferences using RDFS and OWL 2 RL semantics.
- Extracting inferences and expanding the graph/model **programmatically**.
- **Python:** We are using the OWL-RL library
`owlrl.DeductiveClosure(owlrl.RDFS_Semantics).expand(g)`
`owlrl.DeductiveClosure(owlrl.OWLRL_Semantics).expand(g)`

RDFS inference rules (cheatsheet)

Rule	If	Then add	
rdf1	(x p y)	(p rdf:type rdf:Property)	
rdfs2	(p rdfs:domain c) (x p y)	(x rdf:type c)	
rdfs3	(p rdfs:range c) (x p y)	(y rdf:type c)	
rdfs4a	(x p y)	(x rdf:type rdfs:Resource)	
rdfs4b	(x p y)	(y rdf:type rdfs:Resource)	
rdfs5	(p rdfs:subPropertyOf q) (q rdfs:subPropertyOf r)	(p rdfs:subPropertyOf r)	schema only
rdfs6	(p rdf:type rdf:Property)	(P rdfs:subPropertyOf p)	data + schema
rdfs7	(p rdfs:subPropertyOf q) (x p y)	(x q y)	
rdfs8	(c rdf:type rdfs:Class)	(c rdfs:subClassOf rdfs:Resource)	
rdfs9	(c rdfs:subClassOf d) (x rdf:type c)	(x rdf:type d)	
rdfs10	(c rdf:type rdfs:Class)	(c rdfs:subClassOf c)	not relevant
rdfs11	(c rdfs:subClassOf d) (d rdfs:subClassOf e)	(c rdfs:subClassOf e)	
rdfs12	(p rdf:type rdfs:ContainerMembershipProperty)	(p rdfs:subPropertyOf rdfs:Member)	
rdfs13	(x rdf:type rdfs:Datatype)	(x rdfs:subClassOf rdfs:Literal)	
	(p owl:inverseOf q)	(q owl:inverseOf p)	
	(p owl:inverseOf q) (x p y)	(y q x)	
	(p rdf:type owl:SymmetricProperty) (x p y)	(y p x)	

(*) Figure adapted from “Towards Efficient Schema-Enhanced Pattern Matching over RDF Data Streams”. ISWC’11 slides.